Plasticity (described with scalars)

Youngung Jeong

Linear isotropic elasticity

Elastic constitutive law (Hooke's law):

$$\mathbb{E}_{ijkl}\varepsilon_{kl} = \sigma_{ij} \qquad (linear\ elasticity)$$

$$\mathbb{E}_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) \qquad (isotropic\ elasticity;\ two\ constants\ \lambda, \mu)$$

Replacing $\mathbb{E}_{ijkl}$ to the Hooke's law

$$\sigma_{ij} = \mathbb{E}_{ijkl}\varepsilon_{kl} = \lambda\delta_{ij}\delta_{kl}\varepsilon_{kl} + \mu\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right)\varepsilon_{kl}$$
$$= \lambda\delta_{ij}\varepsilon_{kk} + \mu\left(\delta_{ik}\delta_{jl}\varepsilon_{kl} + \delta_{il}\delta_{jk}\varepsilon_{kl}\right) = \lambda\delta_{ij}\varepsilon_{kk} + \mu\left(\delta_{ik}\varepsilon_{kj} + \delta_{il}\varepsilon_{jl}\right)$$
$$= \lambda\delta_{ij}\varepsilon_{kk} + \mu\left(\varepsilon_{ij} + \varepsilon_{ji}\right) = \lambda\delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}$$

What makes plasticity complicated

- Plasticity is more complicated than elasticity, in that constitutive model is not ‘linear’.

If it were linear, as in the elasticity, the constitutive description based on tensors can be expressed as linear algebraic expression such that

$$[\boldsymbol{\sigma}] = [\mathbb{E}][\boldsymbol{\varepsilon}^{el}]$$

- While elasticity deformation occurs whenever $\boldsymbol{\sigma} \neq \mathbf{0}$ is applied. In elasticity, there’s certain criterion which should be met for plasticity to kick in. (Yield criterion, yield surface)

- The non-linearity in plastic constitutive law, is ’differential equation’.

$$\frac{d\varepsilon_{ij}}{dt} = \frac{d\lambda}{dt}\frac{\partial\phi(\boldsymbol{\sigma})}{\partial\sigma_{ij}}$$

$$(\text{two free indices implied: } i, j; \text{Since } \boldsymbol{\sigma} \text{ and } \boldsymbol{\varepsilon} \text{ are symmetric, only 6 equations are implied.})$$

- Strain-hardening occurs.

What makes plasticity complicated

- Elastic behavior of metals (and of most materials) is very well understood. However, plasticity remains relatively not well understood and behaviors are often described empirically (assumptions based on observing the phenomena/experimental behaviors)

- Not a single rule prevails .. (as opposed to Hooke's law for elasticity). There are various mathematical models (theories) developed to describe plastic behavior of metals.

- Experimental methods often are limited to 'uniaxial' cases (where only a single component of stress tensor is not zero). Therefore, with using the available experimental data, it feels like using a 'scalar' clue to describe 'vectorial' (or tensorial) world…

Concept of yield criterion.

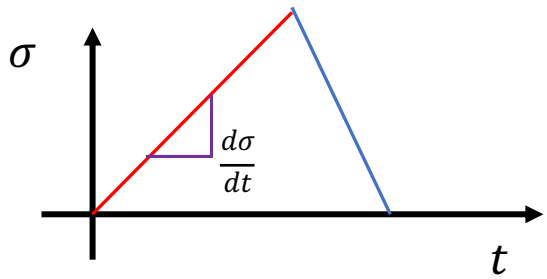
- We introduce a yield criterion as a ***<u>scalar</u>*** function (often denoted as ϕ) of **<u>stress tensor</u>** $\boldsymbol{\sigma}$

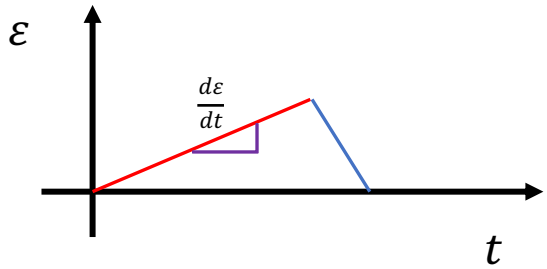
- $\phi = \phi(\boldsymbol{\sigma})$. Just like $f = f(x)$. Both functions return 'scalar' quantity (often denoted by its own symbol such as ϕ or f). The only difference is that the argument is 'tensor' not scalar.

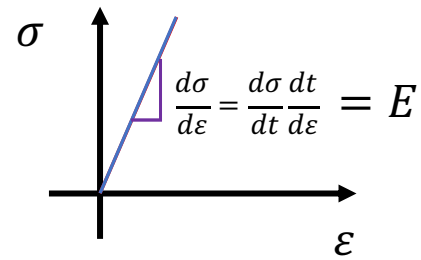
- We say if the stress state of material meets the yield condition if $\phi = c$ with c being the given material characteristic (often representing yield strength)

- We also use this yield function as plastic potential in flow rule (associated flow rule)

Linear elasticity: Behavior of material that exhibits only linear elasticity







Our target is to find the relationship between σ and ε

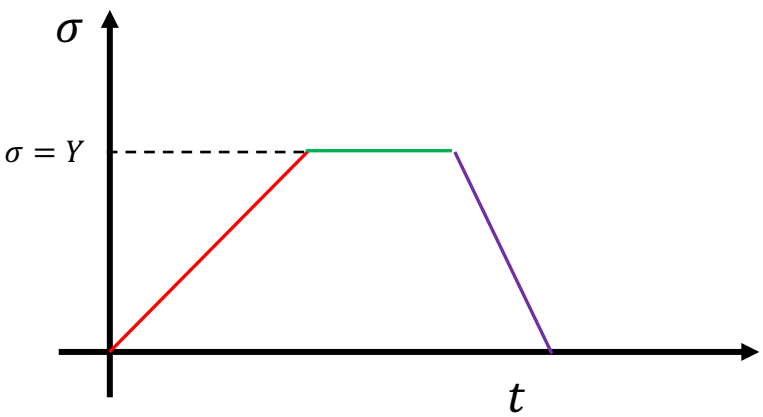
$$\frac{d\sigma}{dt} = E\frac{d\varepsilon}{dt} \quad \rightarrow d\sigma = Ed\varepsilon \quad \rightarrow \int_0^{\sigma} d\sigma = E\int_0^{\varepsilon} d\varepsilon$$

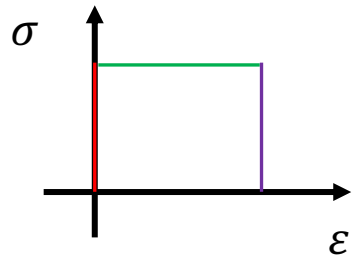
$$\rightarrow \sigma = E\varepsilon$$

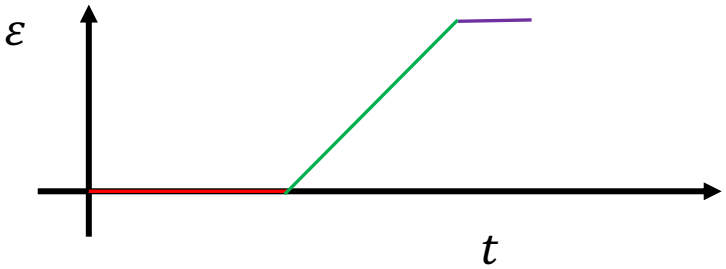
With tensors:

$$\rightarrow \boldsymbol{\sigma} = \mathbb{E}:\boldsymbol{\varepsilon}$$

Rigid perfect plastic







$\frac{d\sigma}{dt} \neq 0, \frac{d\varepsilon}{dt} = 0, \sigma < Y$

$\frac{d\sigma}{dt} = 0, \frac{d\varepsilon}{dt} \neq 0, \sigma = Y$

$\frac{d\sigma}{dt} \neq 0, \frac{d\varepsilon}{dt} = 0, \sigma < Y$

$\frac{d\varepsilon}{dt} = 0, \sigma < Y$

$\frac{d\varepsilon}{dt} = c \ , \sigma = Y$

If $\sigma = Y$ is met , or not → yield criterion

Constitutive model

$$\frac{d\varepsilon}{dt} = c\widehat{H}(\sigma - Y)$$

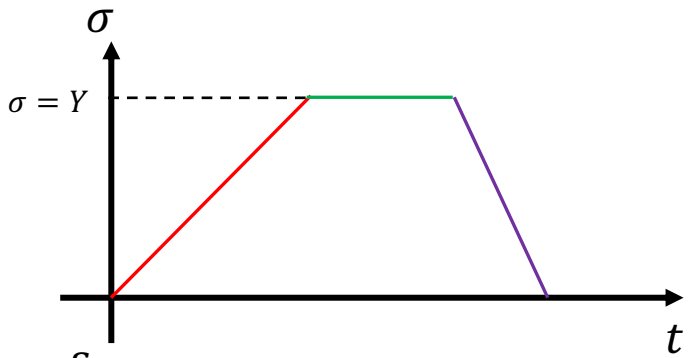
H: Heaviside function;

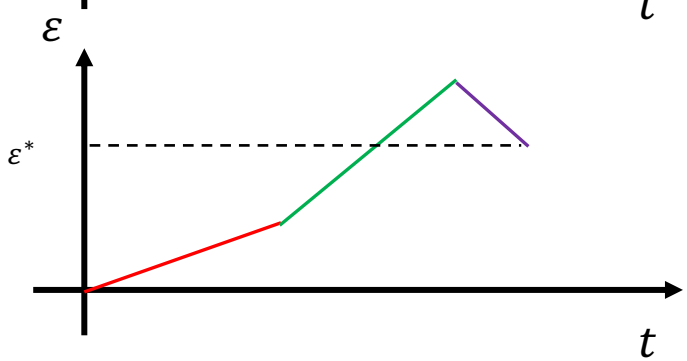
$$H(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

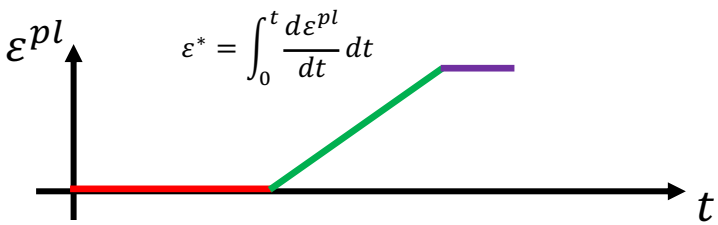
Similarly, Let's define $\widehat{H}$

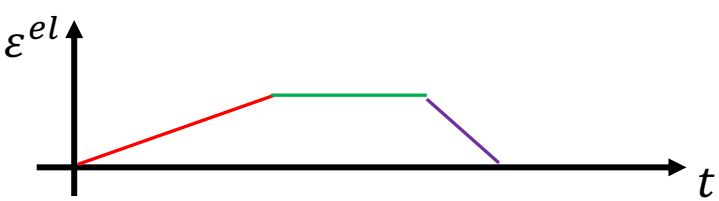
$$\widehat{H}(\sigma - Y) \begin{cases} 1, \sigma = Y \\ 0, \sigma < Y \end{cases}$$

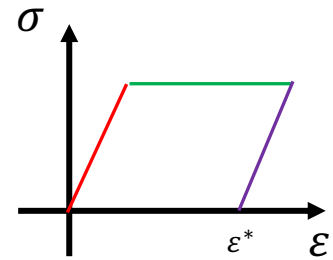
Elastic-Rigid perfect plastic











$$\frac{d\sigma}{dt} \neq 0, \frac{d\varepsilon}{dt} \neq 0, \left(\frac{d\sigma}{dt} = E\frac{d\varepsilon}{dt}\right)\sigma < Y \longrightarrow \frac{d\varepsilon}{dt} = \frac{1}{E}\frac{d\sigma}{dt}, \sigma = E\varepsilon, \sigma < Y$$

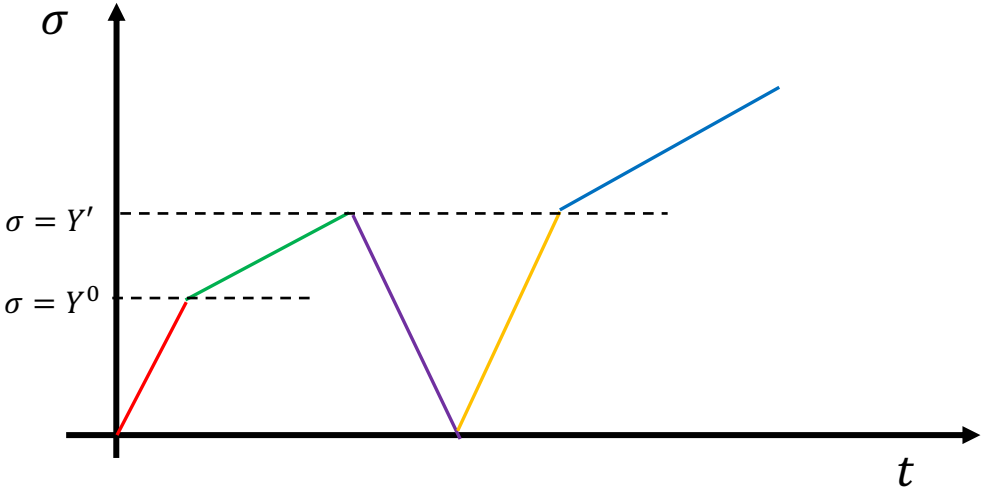
$$\frac{d\sigma}{dt} \neq 0, \frac{d\varepsilon}{dt} \neq 0, \left(\frac{d\sigma}{dt} = E\frac{d\varepsilon}{dt}\right)\ \sigma < Y$$

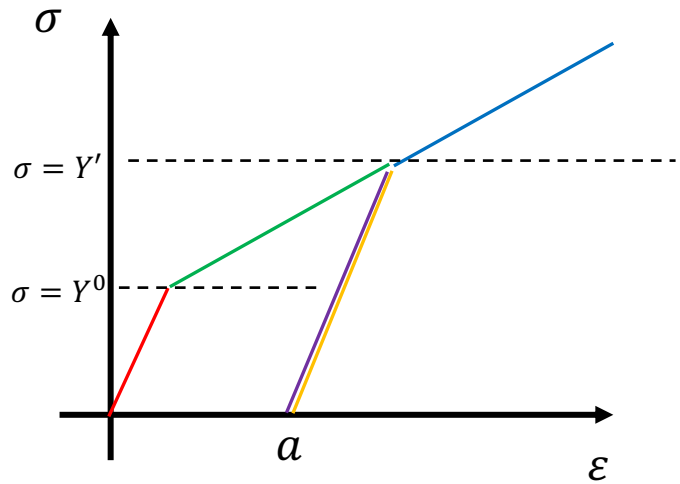
$$\frac{d\sigma}{dt} = 0, \frac{d\varepsilon}{dt} \neq 0, \sigma = Y \qquad \frac{d\varepsilon}{dt} = c, \sigma = Y$$

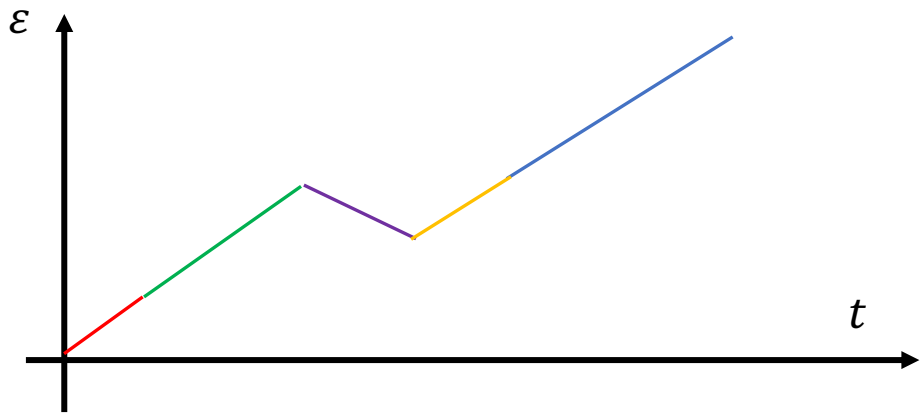
$$\frac{d\varepsilon}{dt} = \frac{d\varepsilon^{pl}}{dt} + \frac{d\varepsilon^{el}}{dt}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}(\sigma - Y) + \frac{1}{E}\frac{d\sigma}{dt} \begin{cases} \text{If } \sigma < Y, & \dfrac{d\varepsilon}{dt} = \dfrac{1}{E}\dfrac{d\sigma}{dt} \\ \text{If } \sigma = Y, & \dfrac{d\varepsilon}{dt} = c + \dfrac{1}{E}\dfrac{d\sigma}{dt} \end{cases}$$

Elastic-plastic with hardening



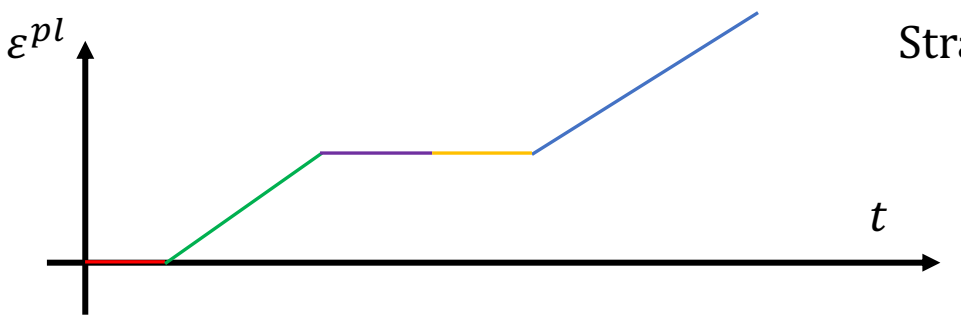




$$\frac{d\varepsilon}{dt} = c\widehat{H}\left(\sigma - \frac{Y' - Y^0}{a}\varepsilon^{pl}(t)\right) + \frac{1}{E}\frac{d\sigma}{dt}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - \frac{Y' - Y^0}{a}\left(\varepsilon(t) - \varepsilon^{el}(t)\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - \frac{Y' - Y^0}{a}\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt}dt\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$



Strain-hardening:

$$Y = Y\left(\varepsilon^{pl}\right) \quad \rightarrow Y = \frac{Y' - Y^0}{a}\varepsilon^{pl}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - k\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt}dt\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$

Simulation using Google spread sheet

You'll need to create your own (custom) function to use $\widehat{H}$.

This site tells you how to create a custom function
https://developers.google.com/apps-script/guides/sheets/functions

I wrote my own as below:

```
function hhat(sig,y) {
  if (sig<y){
    return 0;
  }
  if (sig==y){
    return 1;
  }
  if (sig>y){
    return -1;
  }
}
```

Problem 1)

- Tensile a rigid-plastic body with the initial length of 10 in the rate of

$$\frac{dl}{dt} = 0.001 \left[\frac{mm}{sec}\right]$$

for 10 seconds.

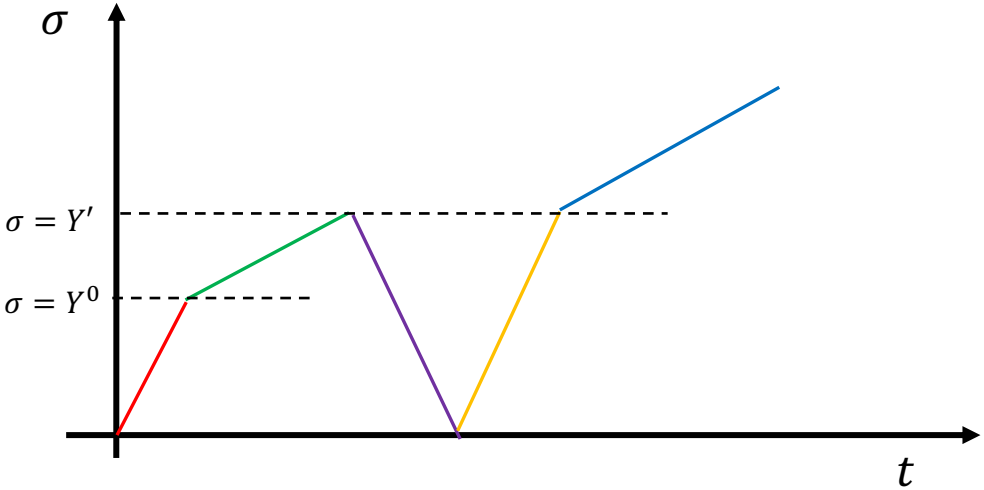
- The body has the elastic modulus of 200,000 MPa and Y as 150 Mpa.

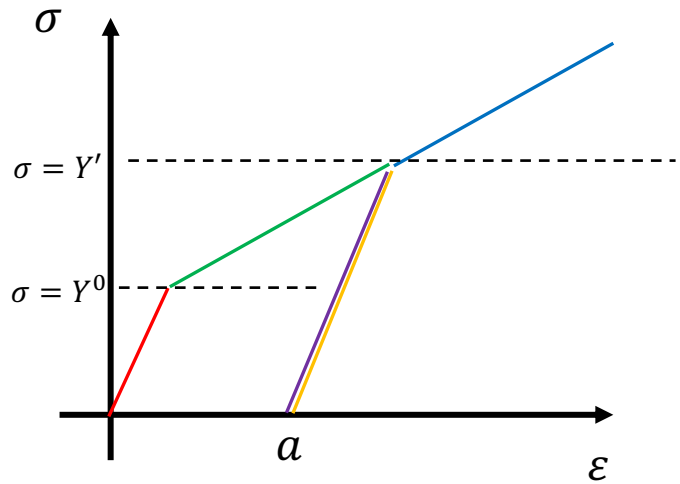
To solve elasto-plastic problem with tensors...

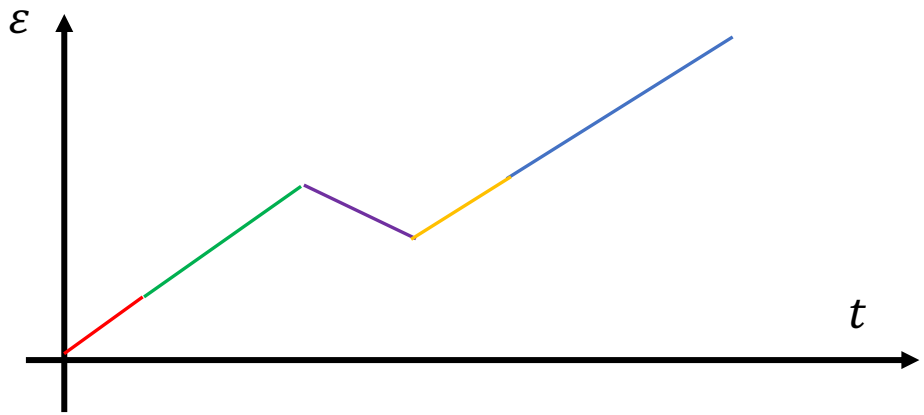
- You'll need a special algorithm

- A most common (and basic) algorithm is 'return-mapping'.
  - 1st you assume that the strain increment is purely elastic, and use Hooke's law to obtain stress increment.
  - If the stress + stress increment does not meet yield criterion, you are safe with this assumption and you are all set. (fin.)
  - If the stress + stress increment gives stress level that is higher than 'yield surface', your assumption is wrong and re-guess elastic strain increment (thus plastic strain increment).
  - For this one, you'll need a numerical iteration (such as Newton-Raphson).
- There are various other algorithms.

Elastic-plastic with hardening



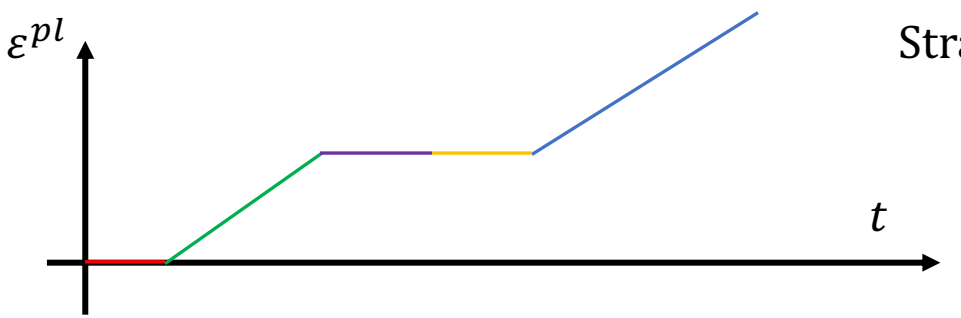




$$\frac{d\varepsilon}{dt} = c\widehat{H}\left(\sigma - \frac{Y' - Y^0}{a}\varepsilon^{pl}(t)\right) + \frac{1}{E}\frac{d\sigma}{dt}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - \frac{Y' - Y^0}{a}\left(\varepsilon(t) - \varepsilon^{el}(t)\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - \frac{Y' - Y^0}{a}\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt}dt\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$



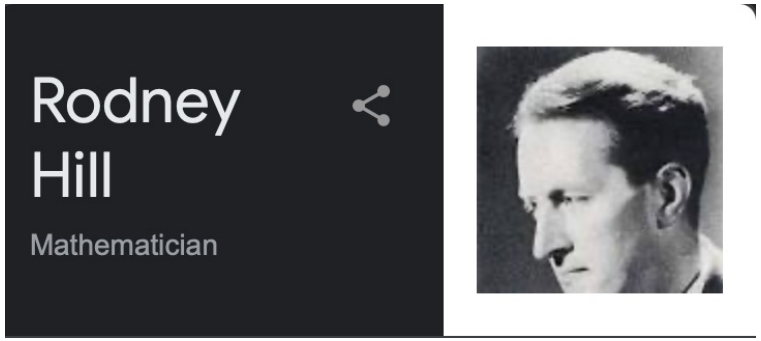
Strain-hardening:

$$Y = Y\left(\varepsilon^{pl}\right) \quad \rightarrow Y = \frac{Y' - Y^0}{a}\varepsilon^{pl}$$

$$\frac{d\varepsilon}{dt} = c\widehat{H}\left\{\sigma - k\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt}dt\right)\right\} + \frac{1}{E}\frac{d\sigma}{dt}$$

Theory of plasticity

- Rodney Hill



Yield criterion with stress tensor (not scalar)

- In order to use tensorial quantities and apply the former method, we'll need to adjust a few assumptions.
- The use of Heaviside like function for yield criterion

$$\widehat{H}(\sigma - Y) \rightarrow \widehat{H}(\boldsymbol{\sigma}, Y). \qquad \widehat{H}\left(\sigma_{ij}, Y\right).$$

We cannot just subtract $\sigma_{ij} - Y$: σ_{ij} is a tensor, Y is a scalar quantity.

We introduce a scalar function, called the yield function.
Yield criterion is described as a function of σ_{ij} (two free indices)

$\phi = \phi\left(\sigma_{ij}\right)$ and Let's use ϕ to see if yield condition is met ($\phi = Y$) or not ($\phi < Y$).

Strain-hardening with stress, strain tensors (not scalars)

- One use length of vector to quantify the 'size' of a vector.
- Similarly, we use equivalent scalar quantities for stress and strain tensors.

- The equivalent scalar quantity for stress tensor is simply called 'equivalent stress', and the same is applied to strain tensor ('equivalent strain').

- There are a few types of equivalent quantities. We'll use only von Mises quantity.

$$\bar{\sigma}^{eq} = \left[\frac{1}{2}\{(\sigma_{11}-\sigma_{22})^2 + (\sigma_{22}-\sigma_{33})^2 + (\sigma_{33}-\sigma_{11})^2 + 6(\sigma_{12}^2+\sigma_{23}^2+\sigma_{13}^2)\}\right]^{0.5}$$

- The definition of equivalent strain is not straightforward. We'll have to determine the amount of plastic work done to the material that is under the given stress σ_{ij} (thus providing us the above equivalent stress tensor).

Plastic work done

- Apparently, plasticity is non-linear, the work done for the time from 0 to t is defined using integration:
- $w^{pl}(t) = w^{pl}(t=0) + \int_0^t dw$
- $dw^{pl} = \boldsymbol{\sigma}: d\boldsymbol{\varepsilon}^{pl} = \sigma_{ij} d\varepsilon_{ij}^{pl}$ (two dummy indices i, j)
- We postulate the work done calculated by stress and strain tensor (as done above) should be the same as the one calculated by the equivalent stress, equivalent strain.
- The above postulation is expressed as below:
- $dw^{pl} = \sigma_{ij} d\varepsilon_{ij}^{pl} = \sigma^{eq} d\varepsilon^{eq} \rightarrow d\varepsilon^{eq} = \frac{\sigma_{ij} d\varepsilon_{ij}^{pl}}{d\sigma^{eq}}$

$$\varepsilon^{eq}(t) = \varepsilon^{eq}(t=0) + \int_0^t d\varepsilon^{eq} = \varepsilon^{eq}(t=0) + \int_0^t \frac{\sigma_{ij} d\varepsilon_{ij}^{pl}}{d\sigma^{eq}}$$

- For materials without previous deformation, the above can be written as:

$$\varepsilon^{eq}(t) = \int_0^t \frac{\boldsymbol{\sigma}: d\boldsymbol{\varepsilon}^{pl}}{d\sigma^{eq}}$$

Strain hardening?

- Say, your material obeys the Hollomon equation,

$$\sigma = K'\varepsilon^n$$

$\boldsymbol{\sigma} = K'\boldsymbol{\varepsilon}^n$ (not possibly); The mathematical operation of 'exponential' is not available.

$\sigma^{eq} = K'(\varepsilon^{eq})^n$; Instead of using the tensors, we use 'equivalent' quantities (that are scalars).

- In case your material obeys linear hardening:

$$\sigma^{eq} = K\varepsilon^{eq}$$

Now, let's look at the constitutive model again

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{c\hat{H}\left\{\boldsymbol{\sigma} - k\left(\boldsymbol{\varepsilon}(t) - \int_0^t \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt} dt\right)\right\}} + \frac{1}{E} : \frac{d\boldsymbol{\sigma}}{dt}$$

Linear strain hardening: $\sigma^{eq} = K'\varepsilon^{eq}$ should somehow replace $k\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt} dt\right)$; Since yield stress is 'strengthened' by plastic deformation

$$\sigma - k\left(\varepsilon(t) - \int_0^t \frac{1}{E}\frac{d\sigma}{dt} dt\right) \quad \rightarrow \quad \phi(\boldsymbol{\sigma}) - \sigma^{eq}(\varepsilon^{eq}) \qquad \varepsilon^{eq}(t) = \int_0^t \frac{\boldsymbol{\sigma} : d\boldsymbol{\varepsilon}}{d\sigma^{eq}}$$

c should be somehow replaced with a tensorial quantity similar to $\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}$, for now, Let's replace c with $\frac{\mathrm{d}\lambda}{dt}\left(\boldsymbol{\varepsilon}^{pl}\right)$ and $d\lambda$ is differential form of a scalar quantity that is a function of plastic strain $\boldsymbol{\varepsilon}^{pl}$.

Now, let's look at the constitutive model again

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{c\widehat{H}\left\{\boldsymbol{\sigma} - k\left(\boldsymbol{\varepsilon}(t) - \int_0^t \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt} dt\right)\right\}} + \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt}$$

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}\widehat{H}(\phi(\boldsymbol{\sigma}) - K'\varepsilon^{eq})} + \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt}$$

Linear strain hardening: $\sigma^{eq} = K'\varepsilon^{eq}$ should somehow replace $k\left(\boldsymbol{\varepsilon}(t) - \int_0^t \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt} dt\right)$;
Since yield stress is 'strengthened' by plastic deformation

$$\boldsymbol{\sigma} - k\left(\boldsymbol{\varepsilon}(t) - \int_0^t \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt} dt\right) \quad \rightarrow \quad \phi(\boldsymbol{\sigma}) - \sigma^{eq}(\varepsilon^{eq}) \qquad \varepsilon^{eq}(t) = \int_0^t \frac{\boldsymbol{\sigma} : d\boldsymbol{\varepsilon}}{d\sigma^{eq}}$$

c should be somehow replaced with a tensorial quantity similar to $\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}$, for now,
Let's replace c with $\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}$ as $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{pl} + \boldsymbol{\varepsilon}^{el}$. $\left(\rightarrow \frac{d\boldsymbol{\varepsilon}}{dt} = \frac{d\boldsymbol{\varepsilon}^{pl}}{dt} + \frac{d\boldsymbol{\varepsilon}^{el}}{dt}\right)$

Finding unknown plastic strain

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}\widehat{H}(\phi(\boldsymbol{\sigma}) - K'\varepsilon^{eq}) + \frac{1}{\mathbb{E}}:\frac{d\boldsymbol{\sigma}}{dt}}$$

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \frac{d\boldsymbol{\varepsilon}^{pl}}{dt}\widehat{H}(\phi(\boldsymbol{\sigma}) - K'\varepsilon^{eq}) + \frac{1}{\mathbb{E}}:\frac{d\boldsymbol{\sigma}}{dt}$$

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \frac{d\boldsymbol{\varepsilon}^{pl}}{dt} + \frac{1}{\mathbb{E}}:\frac{d\boldsymbol{\sigma}}{dt}$$

$$\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}??$$

Let's postulate Hill's idea (associated flow rule)

$$\frac{d\boldsymbol{\varepsilon}^{pl}}{dt} = \frac{d\varepsilon^{eq}}{dt}\left(\frac{\partial\phi(\boldsymbol{\sigma})}{\partial\boldsymbol{\sigma}}\right)$$

If we use von Mises yield criterion:

$$\frac{d\varepsilon_{ij}^{pl}}{dt} = \frac{d\varepsilon^{eq}}{dt}\left(\frac{\partial\left\{\left[\frac{1}{2}\{(\sigma_{11}-\sigma_{22})^2 + (\sigma_{22}-\sigma_{33})^2 + (\sigma_{33}-\sigma_{11})^2 + 6(\sigma_{12}^2+\sigma_{23}^2+\sigma_{13}^2)\}\right]^{0.5}\right\}}{\partial\sigma_{ij}}\right)$$

If we use von Mises yield criterion:

$$\frac{d\varepsilon_{ij}^{pl}}{dt} = \frac{d\left\{\int_0^t \frac{\boldsymbol{\sigma}:d\boldsymbol{\varepsilon}}{d\sigma^{eq}}\right\}}{dt}\left(\frac{\partial\left\{\left[\frac{1}{2}\{(\sigma_{11}-\sigma_{22})^2 + (\sigma_{22}-\sigma_{33})^2 + (\sigma_{33}-\sigma_{11})^2 + 6(\sigma_{12}^2+\sigma_{23}^2+\sigma_{13}^2)\}\right]^{0.5}\right\}}{\partial\sigma_{ij}}\right)$$

Yet, another example with scalars

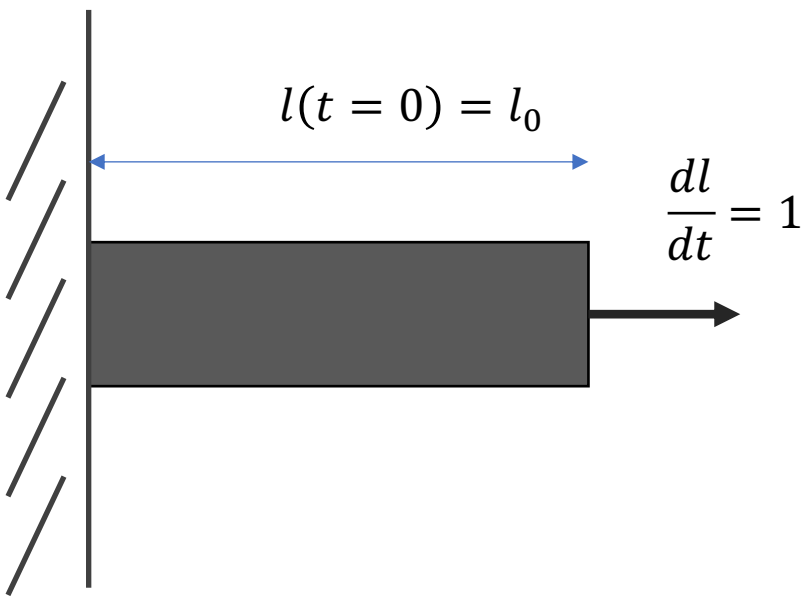
If you are not yet familiar with vector (tensor) algebra and numerical analysis, just feel how much mathematics are involved …

Problem: stretching an elasto-perfect-plastic rod

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}\widehat{H}(\phi(\boldsymbol{\sigma}) - Y^0) + \frac{1}{\mathbb{E}}:\frac{d\boldsymbol{\sigma}}{dt}}$$

Perfect-plastic (no hardening; Y^0 is constant)

with associated flow rule: $\frac{d\varepsilon^{eq}}{dt}\left(\frac{\partial\phi(\boldsymbol{\sigma})}{\partial\boldsymbol{\sigma}}\right)$



For the case of the current 'scalar' example,

$$\varepsilon^{eq} = \varepsilon, \sigma^{eq} = \sigma$$

Let's say, your yield function $\phi(\sigma)$ is

$$\phi(\sigma) = \sqrt{\sigma^2}$$

Your yield criterion:

$$\phi(\sigma) = \sqrt{\sigma^2} = 100$$

Say, your material yield property, $Y^0 = 100$

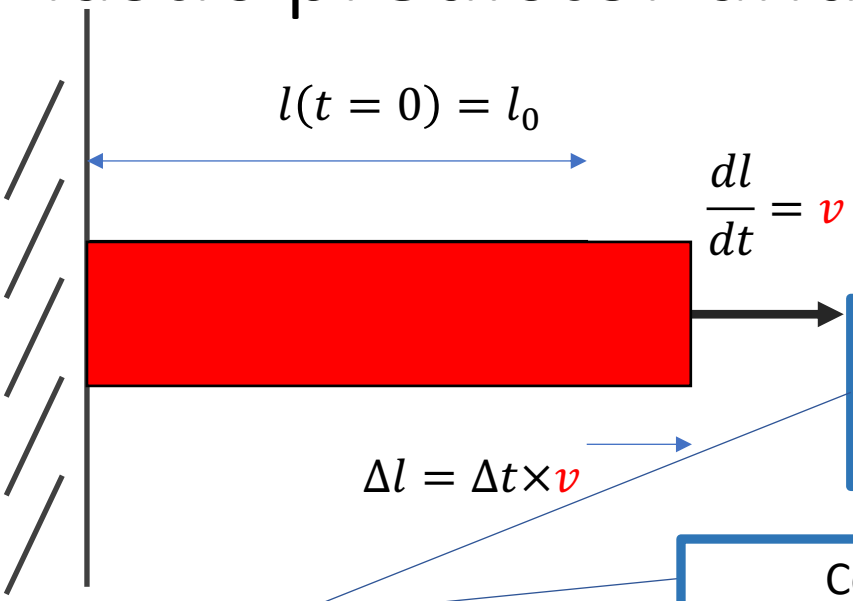
Your yield function gives

$$\frac{d\phi(\sigma)}{d\sigma} = 1$$

and

$\phi(\sigma = 100) = 100$ satisfies yield condition

Elastic predictor and corrector algorithm



While time increment is Δt

$$\Delta\varepsilon=\frac{\Delta l}{l}$$

Initially assuming that all strain is elastic

$$\Delta\sigma=\mathbb{E}\Delta\varepsilon=\mathbb{E}\frac{\Delta l}{l}$$

Correcting elastic strain $\Delta\sigma=\mathbb{E}(\Delta\varepsilon-\Delta\varepsilon^{pl})$

Stress is updated:

$$\sigma_{(n+1)}=\sigma_{(n)}+\Delta\sigma$$
$$\sigma_{(n+1)}=\sigma_{(n)}+\mathbb{E}\Delta\varepsilon$$
$$=\sigma_{(n)}+\mathbb{E}\frac{\Delta l}{l}$$

Check if $\sigma_{(n+1)}$ really gives only elastic contribution

$\phi(\sigma_{(n+1)})<c$ or $\phi(\sigma_{(n+1)})=c$ or $\phi(\sigma_{(n+1)})>c$

You are correct about the strain decomposition, let's move on to next time increment (end)

Plasticity update

$$\Delta\varepsilon^{pl}=\Delta\varepsilon^{eq}\frac{\partial\phi}{\partial\sigma}$$
$$\Delta\varepsilon=\Delta\varepsilon^{el}+\Delta\varepsilon^{pl}$$

Elastic predictor and corrector algorithm

$$\Delta l = \Delta t \times v$$

Initially assuming that all strain is elastic

$$\Delta\sigma = \mathbb{E}\Delta\varepsilon = \mathbb{E}\frac{\Delta l}{l}$$

Find the stress corresponding to the current stress increment

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Correcting elastic strain

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That gives new stress increment

$$\Delta\sigma = \mathbb{E}(\Delta\varepsilon - \Delta\varepsilon^{pl})$$

(adjust strain decomposition

Check if $\sigma_{(n+1)}$ is inside, on or over the yield criterion

$\phi(\sigma_{(n+1)}) < Y^0$ or $\phi(\sigma_{(n+1)}) = Y^0$ or $\phi(\sigma_{(n+1)}) > Y^0$

Plastic strain update

$$\Delta\varepsilon^{pl} = \Delta\varepsilon^{eq}\frac{\partial\phi}{\partial\sigma}$$

If $\phi(\sigma_{(n+1)}) \leq Y^0$

$\sigma_{(n+1)}$ is consistent with our theory.

Let's move on to next time increment

Plastic strain update

$$\Delta\varepsilon^{pl} = \Delta\varepsilon^{eq}\frac{\partial\phi}{\partial\sigma}$$

$F(\Delta\lambda) = \phi(\Delta\varepsilon^{eq}) - (Y^0)$, find $\Delta\varepsilon^{eq}$ that gives $F(\Delta\varepsilon^{eq}) = 0$

That's exactly what NR can do.

$$\frac{dF(\Delta\varepsilon^{eq})}{d\Delta\varepsilon^{eq}} = \frac{d(\phi(\sigma) - Y^0)}{d\Delta\varepsilon^{eq}} = \frac{d\phi(\sigma)}{d\Delta\varepsilon^{eq}} = \frac{d\phi(\sigma)}{d\sigma}\frac{d\sigma}{d\Delta\varepsilon^{eq}} = 1\times\frac{d(\sigma_{(n)} + \Delta\sigma)}{d\Delta\varepsilon^{eq}} = \frac{d\Delta\sigma}{d\Delta\varepsilon^{eq}}$$

$$= \frac{d\mathbb{E}\left(\Delta\varepsilon - \Delta\varepsilon^{pl}\right)}{d\Delta\varepsilon^{eq}} \approx \mathbb{E}\frac{d\Delta\varepsilon^{pl}}{d\Delta\varepsilon^{eq}} = \mathbb{E}\,\mathrm{sgn}\left(\Delta\varepsilon^{pl}\right)$$

$$\Delta\varepsilon^{eq}_{(k+1)} = \Delta\varepsilon^{eq}_{(k)} - \frac{F\left(\Delta\varepsilon^{eq}_{(k)}\right)}{-\sigma\mathbb{E}\,\mathrm{sgn}(\Delta\varepsilon^{pl})}$$

$\mathrm{sgn}\left(\Delta\varepsilon^{pl}\right) = 1$ if $\Delta\varepsilon^{pl} \geq 0$

$\mathrm{sgn}\left(\Delta\varepsilon^{pl}\right) = -1$ if $\Delta\varepsilon^{pl} < 0$

Caution! Need validation

Cheat sheet

```
real function calc_yield_function(stress)
implicit none
real stress
calc_yield_function = sqrt(stress**2.)
return
end function
```

```
program elasto_plasticity_scalar
implicit none
real calc_yield_function
real dt, E, c, t, l, dl, eps, deps, deps_el, deps_pl,dsig,
$     stress
real vel, f, tol
integer kount,iplast
parameter(tol=1e-6)

open(3,file='elasto_plasticity_scalar.txt')
dt = 0.02                    ! time increment
E = 200000                   ! elastic modulus
c = 200.                     ! yield criterion
stress = 0.                  ! initial stress
eps = 0.                     ! initial strain
l = 1.                       ! initial length
t = 0.                       ! initial time
```

```
    do while(t<1.0)
c   Loading condition 1
       if (t.le.0.25) then
          vel = 0.01
       elseif (t.gt.0.25.and.t.le.0.55) then
          vel =-0.01
       else
          vel = 0.01
       endif
       dl = vel * dt
       deps = dl / l
c      initially assuming all strain is elastic
       deps_pl = 0.0
       deps_el = deps
c      guess on stress increment
       dsig = E * deps_el
       kount = 0
       f = calc_yield_function( stress+ dsig) - c
       iplast=0
       do while (f.gt.tol.and.kount.lt.3) ! if exceeding plastic onset
          iplast=1
          deps_el = deps - deps_pl
          dsig = e * deps_el
          f = calc_yield_function(stress+dsig) - c
c         estimate new plastic increment
          deps_pl = deps_pl - f/(-E)*sign(1.,deps)
          kount = kount +1
       enddo
       write(3,'(2f9.4,2f10.4,2i2)')t,l,eps,stress,iplast,kount
       stress = stress + dsig
       eps = eps + deps
       t = t + dt
       l = l + dl
    enddo
    close(3)
    end program
```

Problem: stretching an elasto-plastic rod with linear hardening

$$\frac{d\boldsymbol{\varepsilon}}{dt} = \boxed{\frac{d\boldsymbol{\varepsilon}^{pl}}{dt}\hat{H}(\phi(\boldsymbol{\sigma}) - (Y^0 + k\varepsilon^{eq}))} + \frac{1}{\mathbb{E}} : \frac{d\boldsymbol{\sigma}}{dt}$$

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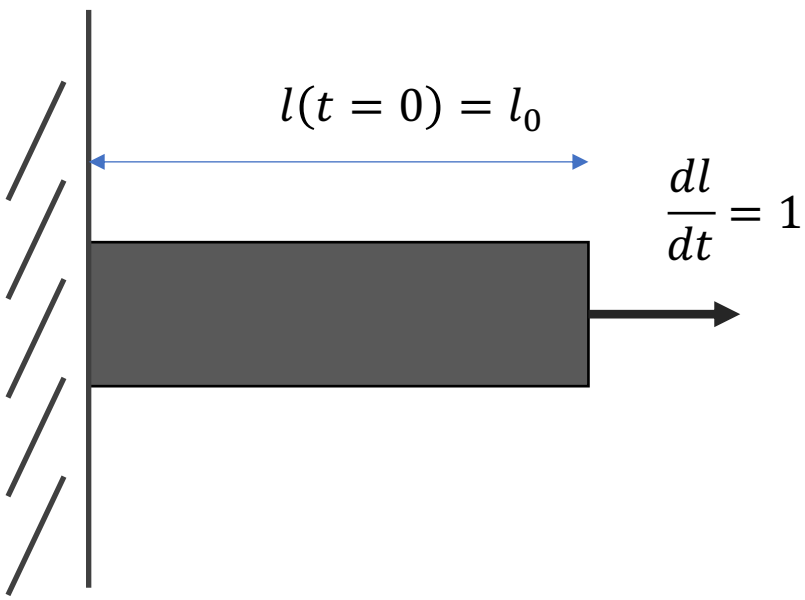
with associated flow rule: $\frac{d\varepsilon^{eq}}{dt}\left(\frac{\partial\phi(\boldsymbol{\sigma})}{\partial\boldsymbol{\sigma}}\right)$



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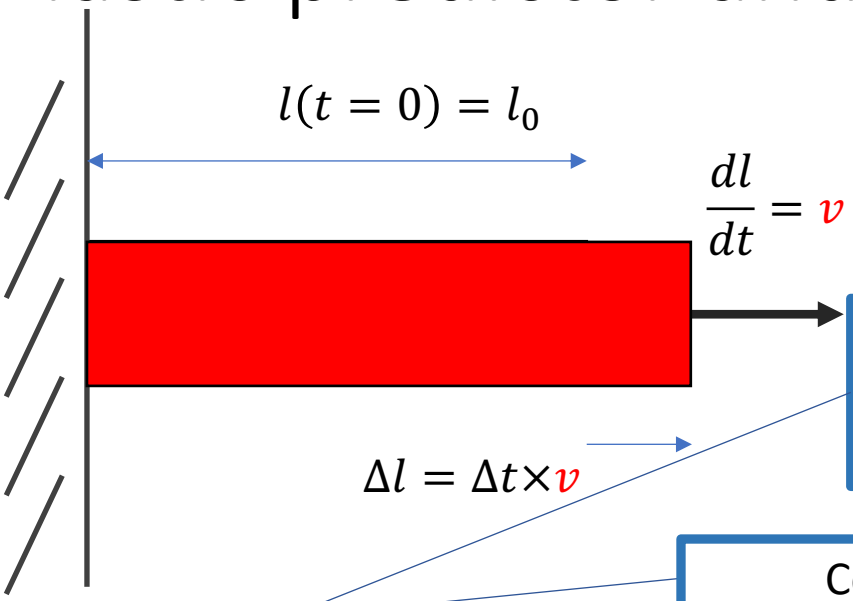

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$$\Delta\sigma = \mathbb{E}(\Delta\varepsilon - \Delta\varepsilon^{pl})$$

(adjust strain decomposition

Check if $\sigma_{(n+1)}$ is inside, on or over the yield criterion

$$\phi\left(\sigma_{(n+1)}\right) < Y^0 + k(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq})$$

or $\phi\left(\sigma_{(n+1)}\right) = Y^0 + k(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq})$

or $\phi\left(\sigma_{(n+1)}\right) > Y^0 + k(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq})$

Plastic strain update

$$\Delta\varepsilon^{pl} = \Delta\varepsilon^{eq}\frac{\partial\phi}{\partial\sigma}$$

If $\phi\left(\sigma_{(n+1)}\right) \leq Y^0 + k(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq})$

$\sigma_{(n+1)}$ is consistent with our theory.

Let's move on to next time increment

Plastic strain update

$$\Delta\varepsilon^{pl} = \Delta\varepsilon^{eq}\frac{\partial\phi}{\partial\sigma}$$

That's exactly what NR can do.

$F(\Delta\lambda) = \phi(\Delta\varepsilon^{eq}) - \left(Y^0 + k\left(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq}\right)\right)$, find $\Delta\varepsilon^{eq}$ that gives $F(\Delta\varepsilon^{eq}) = 0$

$$\frac{dF(\Delta\varepsilon^{eq})}{d\Delta\varepsilon^{eq}} = \frac{d\left(\phi(\sigma) - \left(Y^0 + k\left(\varepsilon^{eq}_{(n)} + \Delta\varepsilon^{eq}\right)\right)\right)}{d\Delta\varepsilon^{eq}} = \frac{d\phi(\sigma)}{d\Delta\varepsilon^{eq}} - k = \frac{d\phi(\sigma)}{d\sigma}\frac{d\sigma}{d\Delta\varepsilon^{eq}} - k$$

$$= 1\times\frac{d(\sigma_{(n)} + \Delta\sigma)}{d\Delta\varepsilon^{eq}} - k = \frac{d\Delta\sigma}{d\Delta\varepsilon^{eq}} - k = \frac{d\mathbb{E}\left(\Delta\varepsilon - \Delta\varepsilon^{pl}\right)}{d\Delta\varepsilon^{eq}} - k \approx \mathbb{E}\frac{d\Delta\varepsilon^{pl}}{d\Delta\varepsilon^{eq}} - k = \mathbb{E}\,\mathrm{sgn}\left(\Delta\varepsilon^{pl}\right) - k$$

$$\Delta\varepsilon^{eq}_{(k+1)} = \Delta\varepsilon^{eq}_{(k)} - \frac{F\left(\Delta\varepsilon^{eq}_{(k)}\right)}{-\sigma\mathbb{E}\,\mathrm{sgn}(\Delta\varepsilon^{pl}) + k}$$

$\mathrm{sgn}\left(\Delta\varepsilon^{pl}\right) = 1$ if $\Delta\varepsilon^{pl} \geq 0$

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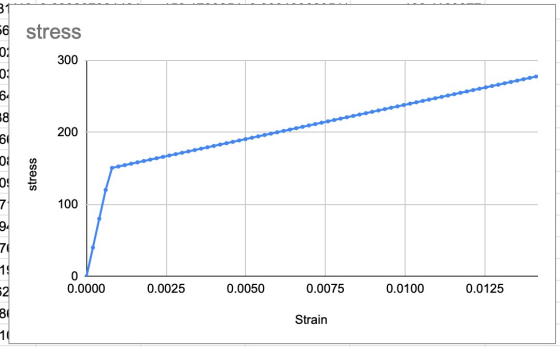
Caution! Need validation

Google spreadsheet example on elasticity-linear plasticity (scalar)

https://docs.google.com/spreadsheets/d/1ithHT-f_mQIyv9Vx2fBt_o6bMWRq8Hlb0N5VXrw2ySA/edit?usp=sharing

my loading condition									Elast Modulus (M
time incr (dt)	0.2		dl/dt	0.01	Delta l	0.002			200,000
time	length	eps	eps_pl	stress	dstrain	dsigma (pure)+stress	H (sig^guss-Y)	deps_elasticity	deps_plas
0	10	0	0	0	0.0002	40	0	0.0002	0
0.2	10.002	0.0002	0	40	0.000199960008	79.9920016	0	0.000199960008	0
0.4	10.004	0.000399960008	0	79.9920016	0.000199920032	119.976008	0	0.000199920032	0
0.6	10.006	0.00059988004	0	119.976008	0.000199880072	159.9520224	-1	0.000152607965	0.000047272106
0.8	10.008	0.000799760111	0.000047272106	150.4976011	0.000199840127	190.4656267	-1	0.000009873826	0.000189966301
1	10.01	0.000999600239	0.000237238408	152.4723663	0.000199800199	192.4324063	-1	0.000009515094	0.000190285105
1.2	10.012	0.00119940044	0.000427523513	154.3753852	0.000199760287	194.3274427	-1	0.000009512301	0.000190247986
1.4	10.014	0.001399160727	0.000617771499	156.2778455	0.000199720391	196.2219238	-1	0.000009510399	0.000190209991
1.6	10.016	0.00159888					-1	0.000009508500	0.000190172010
1.8	10.018	0.00179856					-1	0.000009506602	0.000190134044
2	10.02	0.00199820					-1	0.000009504704	0.000190096093
2.2	10.022	0.00219780					-1	0.000009502808	0.000190058157
2.4	10.024	0.00239736					-1	0.000009500912	0.000190020237
2.6	10.026	0.00259688					-1	0.000009499016	0.000189982331
2.8	10.028	0.00279636					-1	0.000009497122	0.000189944441
3	10.03	0.00299580					-1	0.000009495228	0.000189906566
3.2	10.032	0.00319520					-1	0.000009493335	0.000189868705
3.4	10.034	0.00339457					-1	0.000009491443	0.000189830860
3.6	10.036	0.00359389					-1	0.000009489551	0.000189793030
3.8	10.038	0.00379317					-1	0.000009487661	0.000189755215
4	10.04	0.00399241					-1	0.000009485771	0.000189717415
4.2	10.042	0.00419162					-1	0.000009483882	0.000189679631
4.4	10.044	0.00439078					-1	0.000009481993	0.000189641861
4.6	10.046	0.00458991					-1	0.000009480105	0.000189604106
4.8	10.048	0.004788994471	0.003846186202	188.5616537	0.000199044586	228.3705709	-1	0.000009478219	0.000189566367




```
function decompose_lh(s,E,y,de,eeq,k){
// elasticity - plasticity with linear hardening
// initial guess with assuming pure elasticity
deq=0
del=de-deq
// F=yield(s+E*del)-y
tolerance=1e-9
F=2*tolerance
kount=0
while(F>tolerance){
// calculate F
F=yield(s+E*del)-(y+k*(eeq+deq))
deq=deq-F/(-E)//*sign(1,de)
if (de<0) {
deq=deq*-1
}
del=de-deq
kount=kount+1
if (kount>20){
throw 'too many iterations', kount
}
}
return del
}
```

Or, Jupyter notebook (Python)

In [1]:
```
%pylab inline
```

```
Populating the interactive namespace from numpy and matplotli
b
```

- Yield surface $\phi = \sqrt{\sigma^2}$

In [2]:
```
def ys(s): return np.sqrt(s**2)
```

In [3]:
```
## predictor corrector algorithm to determine plastic strain i
def decompose(s,E,y0,de,eeq,k):
    deq=0
    deel=de-deq
    tolerance=1e-9
    #F=2*tolerance
    F=ys(s+E*deel)-(y0+k*(eeq+deq))
    kount=0
    while F>tolerance:
        F=ys(s+E*deel)-(y0+k*(eeq+deq))
        deq=deq-F/((-E)*np.sign(de))
        deel=de-deq
        kount=kount+1
        #print('deq:',deq)
        if kount>20:
            raise IOError("something went wrong")
    return deel
```

물성

In [4]:
```
E=200000 ## 영률 (Young's modulus)
y0=200   ## 초기 항복 강도
k=10000  ## Hardening parameter
decompose(100,E,y0,1e-3,0,k)
```

Out[4]:
```
0.0005250000000000001
```

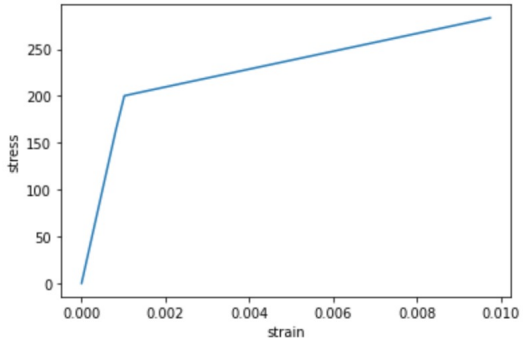
하중 조건 (일축인장)

In [5]:
```
t=np.linspace(0,10) ## 시간 0초 부터 10초 까지
l=np.linspace(10,10.1) ## 길이 10[mm]에서 10.1[mm]까지 변형
```

In [6]:
```
## initially zero stress, zero plastic equivalent strain
s=0
eeq=0
e=0
x=[]
y=[]
for i in range(len(t)-1):
    #x.append(eeq)
    x.append(e)
    y.append(s)
    dl=l[i+1]-l[i]
    de=dl/l[i]
    e=e+de
    deel = decompose(s,E,y0,de,eeq,k)
    s=s+deel*E
    depl=de-deel
    eeq=eeq+depl
```

In [7]:
```
fig=plt.figure();ax=fig.add_subplot(111)
ax.plot(x,y,'-')
ax.set_xlabel('strain')
ax.set_ylabel('stress')
```

Out[7]:
```
Text(0, 0.5, 'stress')
```



https://gist.github.com/youngung/07615666a85d663eae9cbb157fa33e38